Kapasa Makasa University

School of Applied Sciences and Education

ICT Department

Development of a Command Line Interface (CLI) Tool

'sokonalysis' – The Cipher Toolkit Built for All Skill Levels

By:

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Abstract

Cryptographic algorithms are at the heart of digital security. However, applying them remains a challenge to many, particularly beginners in cybersecurity. This project aims to design and implement ‘sokonalysis’, a versatile, user-friendly cipher analysis toolkit that supports both basic and advanced cipher techniques such as Symmetric, Asymmetric, and Hashing algorithms. The toolkit will include features for Capture The Flag (CTF), and Advanced level cryptography, thereby catering to all skill levels. Using a C++ and python backend, and an intuitive Command Line Interface (CLI) for both Linux and Windows Operating System (OS), the toolkit will serve as a practical utility.

This proposal outlines the background, justification, objectives, and methodology that will guide the development of ‘sokonalysis’, including code structure, user interface design, and testing plans. The project is expected to enhance practical cryptanalysis by making cipher techniques more accessible and interactive.

Keywords – *Cryptography*, *Cipher*, *Symmetric*, *Asymmetric*, *Hashing*, *Cybersecurity* and *Capture The Flag (CTF)*, *sokonalysis*

Declaration

Candidate Declaration

I JAMES SOKO declare that this research proposal is my original work and has not been presented for any degree in any other University.

Signature: ……

Date: …….

JAMES SOKO

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List of Abbreviation

AES	Advanced Encryption Standard
API	Application Programming Interface
CLI	Command Line Interface
Crypto++	CryptoPP Library
CTF	Capture The Flag
CTO	CrypTool Online
DES	Data Encryption Standard
GB	Gigabyte
GUI	Graphical User Interface
GMP	Gnu MultiPrecision Library
LLMs	Large Language Models
MinGW	Minimalist GNU for Windows
OpenSSL	Open Secure Sockets Layer Library
OS	Operating System
RSA	Rivest-Sharmir-Adleman
RAM	Random Access Memory
UAT	User Acceptance Testing
VScode	Visual Studio Code

1. CHAPTER ONE: INTRODUCTION

Cryptography is “the art of writing or solving codes”. Cryptography is a Greek word that is a combination of two words cryptos which means “hidden or secret” and graph which means “to write or study.” So, cryptography is well defined as “the study of hidden secrets”. [1] Nowadays, however, the privacy of individuals and organizations is provided through cryptography at a high level, making sure that information is sent securely in a way so that the authorized receiver can access this information. [2]

As threats evolve, so does the need for strong encryption and the skills to analyse it. However, practical tools for analysing ciphers are often either too complex or too limited for most users. This project introduces ‘sokonalysis’, a multi-functional cipher toolkit designed for users at all skill levels.

1.1 Background

With the increasing relevance of cybersecurity, institutions and individuals are adopting encryption tools to protect sensitive data. Common algorithm ciphers like Symmetric, Asymmetric, and Hashing are foundational topics in cryptographic. Unfortunately, many users struggle to understand and apply these techniques due to the abstract nature of the math involved and the lack of practical, beginner-friendly tools.

1.2 Statement of the problem

Many cryptographic tools target advanced audiences with backgrounds in mathematics and programming yet many users have a poor mathematical background. [3] Most CTF crypto challenges require integrating multiple tools and scripting to solve even a single problem. [4]

1.3 Justification

This project is crucial for research in cryptography and cybersecurity. It will enable beginners and professionals to simulate and test ciphers for CTFs and security validation.

1.4 Objectives

1.4.1 General Objective

Developing a Command Line Tool ‘sokonalysis’ the cipher toolkit built for all skill levels.

1.4.2 Specific Objectives

1) To implement basic cryptographic algorithms.
2) To implement common cryptographic algorithms used in CTF competitions.
3) To implement advanced cryptographic services for enterprise and general users.

1.5 Research Questions

1) How effective can ‘sokonalysis’ be in simulating cryptography-related problems?
2) What cryptographic challenges are frequently encountered in CTF competitions?
3) What cryptographic services can be offered to meet the needs of enterprises and general users?

1.6 Scope

The scope includes the design, development, and testing of a standalone cipher analysis toolkit named ‘sokonalysis’. The toolkit will provide support for Symmetric (Stream and Block Ciphers), Asymmetric, and Hashing Algorithms.

Advanced Cryptography and CTF common crypto algorithms for challenges are part of the tool menu, with a dedicated feature that allows users to solve CTF crypto problems without needing to integrate multiple tools.

Unlike existing solutions, ‘sokonalysis’ will include a structured menu system that allows users to navigate through categories directly. The system will employ a structured CLI menu that enables users to navigate through categories directly in a simple, user-friendly manner. The scope is limited to:

1. A CLI implementation only, with no graphical interface.
2. Support for cryptographic CTF challenges only, excluding other categories such as Forensics, Reverse Engineering, or Web Exploitation.

2. CHAPTER TWO: LITERATURE REVIEW

2.1 Challenges of applying Cryptography and Recommended Solution

Many users struggle with cryptography because of its abstract mathematical underpinnings. For example, a weak foundation in number theory and discrete mathematics makes it difficult to grasp concepts such as modular arithmetic, Euler's theorem, and RSA implementation. This is consistent with the findings of Adamović et al., who highlighted the importance of using software tools to support cryptography application [3].

Nielson and Monson further argue that by directly implementing algorithms in Python, users can gradually develop both mathematical intuition and programming competence. This reduces cognitive overload and makes cryptography more approachable, especially for beginners. Their approach demonstrates how a practical, programming-driven methodology can transform cryptography from an abstract subject into a tangible, skill-based discipline [6].

The interactive approach, through practical demonstrations, has been shown to be advantageous. It fosters experimental work and encourages users to engage in discussions to resolve problems. The use of interactive software helps to increase users' interest in cryptography, making the subject more accessible and understandable, especially for those with weaker mathematical foundations.

Similarly, Nielson and Monson emphasize in Practical Cryptography in Python that real-world, hands-on exercises significantly improve users' engagement and understanding by bridging the gap between mathematical theory and its practical applications in programming environments [6].

2.2 Need for Real-World Crypto Application through CTFs

CrypTool is free, open-source Windows GUI software designed for cryptography. It has been under development since 1998 and currently many people are working worldwide on its improvements and maintenance. Its catalogue of state-of-the-art cryptographic functions is nearly comprehensive. It provides resources for both encryption and decryption and coverage from ancient times to present. Features for cryptography include Caesar, Vigenère, Hill, Playfair, transposition, permutation, DES, AES and RSA.

While CrypTool has been praised for its broad coverage, Nielson and Monson point out that tools without a strong programming-based learning component can leave users unprepared for real-world cryptographic challenges. [6].

Capture The Flag (CTF) competitions are increasingly recognized as effective environments for applying cryptographic knowledge. These challenges often require the integration of multiple tools and problem-solving approaches. Existing tools, however, are either too advanced or lack the necessary functionality for beginners. ‘Sokonalysis’ seeks to address this limitation by creating a unified environment that supports practical cryptanalysis for CTFs.

Nielson and Monson highlight the importance of coding practice for preparing users for such real-world cryptographic problem-solving scenarios. Their emphasis on solving challenges “by example” resonates with the CTF learning model, where participants must apply theory to novel, often adversarial situations [6].

According to AICrypto, CTF challenges emphasize practical skills, requiring models to exploit vulnerabilities through source code analysis and numerical reasoning, mimicking real world cryptographic attacks. Their CTF section comprises 150 challenges across eight categories, with 137 taken from competitions held after 2023 to ensure relevance, and the remaining 13 drawn from earlier events. CTF challenges closely mirror real-world attack scenarios and demand practical exploitation skills. While CTF-style tasks have been adopted to evaluate LLMs’ cybersecurity capabilities, prior efforts are limited by a narrow selection of crypto-focused challenges and inconsistent quality standards [7].

AICrypto lack full evaluation automation, therefore, we rely on human experts to evaluate such problems. Nonetheless, developing reliable automated evaluation methods could be an open challenge and yet ‘sokonalysis’ is the game changer, because it does not rely on past events but known attacks and the algorithm behind it.

3. CHAPTER THREE: METHODOLOGY

This project will adopt the Waterfall Software Development Model. The Waterfall model is a linear, sequential approach that divides the system development process into distinct phases. Each phase must be completed before the next begins, ensuring a structured and disciplined methodology. The phases are: Requirements Analysis, System Design, Implementation, Testing, Deployment, and Maintenance.

3.1 System Development Life Cycle (SDLC) Diagram

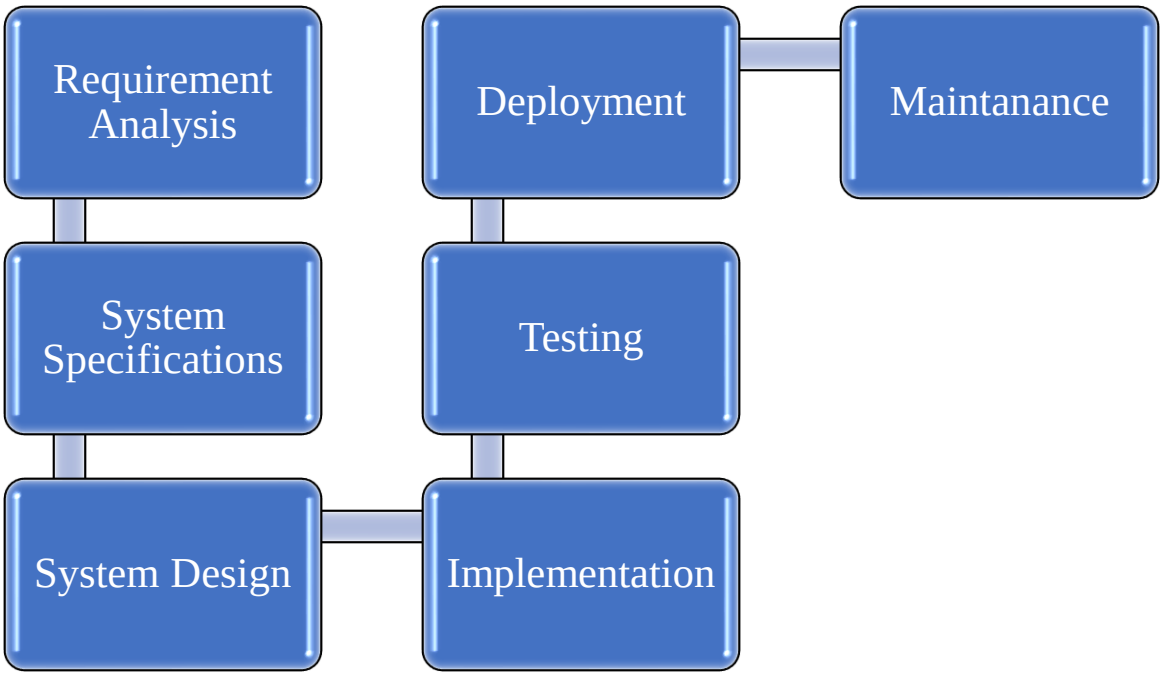

Figure 1:SDLC Diagram

3.2 Requirements Analysis

Gather functional and non-functional requirements through surveys and literature review. An ongoing survey with over 40 responses has been conducted to understand user challenges in cryptography (see **Appendix A: Survey Questionnaire**). All survey responses will be collected voluntarily and kept confidential. The data will only be used for academic purposes, and no personal information will be disclosed.

3.2.1 System Specifications

3.2.1.1 For Developers (Development Environment)

Hardware: Minimum Intel i5 processor, 8GB RAM, 512GB storage.

Software: Operating Systems: Windows 8/10/11 and Linux distributions (Ubuntu, Parrot OS, Kali Linux).

3.2.1.2 For Users (Execution Environment)

Hardware: Minimum Intel i3 processor, 4GB RAM, 128GB storage.

Software: Operating Systems: Windows 8/10/11 or Linux (Ubuntu, Parrot OS, Kali Linux).

3.3 System Design

Develop the overall system architecture using modular design principles. Classify toolkit features into three main levels: Basic Level – classical ciphers (Caesar, Vigenère, Hill, etc.). CTF Level – cryptographic challenges (RSA attacks, AES attacks, Diffie-Hellman attacks, etc.). Advanced Level – modern algorithms.

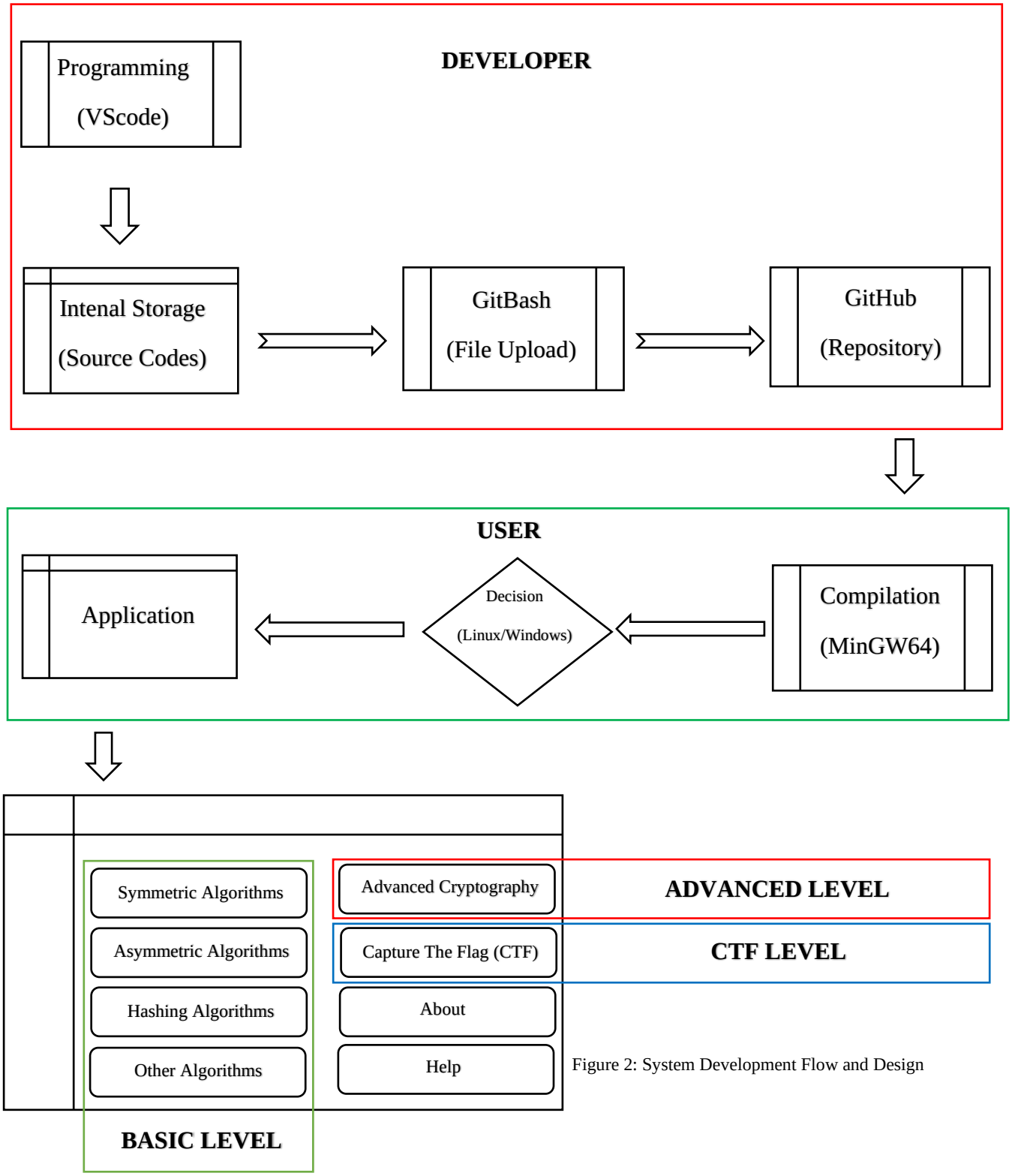

Figure 2: System Development Flow and Design

3.4 Implementation

Use C/C++ for core cryptographic logic and Python for complex mathematical operations. Employ libraries such as Crypto++, OpenSSL, gmpxx, and nlohmann/json and modules cryptography, hashlib, os, PyCryptodome, colorama, sympy, and gmpy2. Apply GitHub for version control and collaborative development. Build the Command Line Interface (CLI) with colorized menus and intuitive navigation.

3.5 Testing

Perform unit testing for individual cipher functions. Conduct integration testing to ensure modules interact correctly. Execute system testing to validate toolkit performance on Windows and Linux. Carry out user acceptance testing (UAT) by engaging CTF players to provide feedback.

3.6 Deployment

Deploy the final version on multiple platforms: GitHub, and SourceForge. Provide installation instructions and user manuals. All cryptographic functions are for educational use. No malicious use is promoted. Users will agree to terms on the first launch. Conduct training sessions and demonstrations for students and lecturers.

3.7 Maintenance

Continuously monitor feedback from users for improvement. Release periodic updates to add new cryptographic algorithms and fix bugs. Ensure long-term sustainability by maintaining the GitHub repository.

4. Bibliography

[1] V. R and S. A. K, "Cryptography: Avalanche effect of AES and RSA," *International Journal of Scientific and Research Publications,* pp. 119-125, 2020.

[2] S. N, P. and K. H, "A Review of Information Security Using Cryptography Technique," *International Journal of Advanced Research in Computer Science,* vol. VIII, pp. 323-326, 2017.

[3] S. Ž. Adamović, M. Šarac, D. Stamenković and D. Radovanovic, "The Importance of the Using Software Tools for Learning Modern Cryptography," *International Journal of Engineering Education,* vol. XXXIV, pp. 256-262, 2018.

[4] S. U. Bhuiyan and L. Ali, "Crypto-master: A Computer Aided Learning Tool for Cryptography," *International Conference on Innovation in Engineering and Technology (ICIET),* pp. 1-5, 2018.

[5] M. Mehreen, Usability Analysis of CrypTool-Online and CrypTool 2.

[6] S. J. Nielson and C. K. Monson, Practical Cryptography in Python: Learning Correct Cryptography by Example, New York: Apress, 2019.

[7] Y. Wang, Y. Liu, L. Ji, H. Luo, . W. Li, X. Zhou, R. Xu, Y. Chen and T. He, AICRYPTO: A COMPREHENSIVE BENCHMARK FOR EVALUATING CRYPTOGRAPHY CAPABILITIES OF LARGE LANGUAGE MODELS, Beijing, 2025.

[8] J. Sequeira and S. Ghazi, OpenSSL, 2019.

[9] J. L. Villar, Basics of OpenSSL, 2024.

[10] T. Kahlert and K. Giza , Visual Studio Code Tips & Tricks, 1st ed., M. Schiffer, Ed., Microsoft, 2016.

[11] D. Stenberg, "The library," in *Everything curl*, 2019, pp. 8-9.

[12] R. Grimes, " Libraries," in *Beginning C++ Programmin*, Birmingham - Mumbai, Packt Publishing Ltd., 2017, pp. 34-35.

5. WORK PLAN

Project Activities (2025 – 2026)

Activity	Start	Finish	Duration (Days)
Identify research field & crypto journals for gap	19/04/2025	15/05/2025	26
Identify problem & set title	16/05/2025	19/06/2025	34
Plan project design	20/06/2025	10/07/2025	20
Prepare survey	11/07/2025	16/07/2025	5
Write synopsis	17/07/2025	27/07/2025	10
Proposal writing & defense	28/07/2025	31/08/2025	34
System design development	20/09/2025	04/10/2025	14
Implement basic algorithms	05/10/2025	20/12/2025	76
Implement common CTF algorithms	21/12/2025	27/05/2026	157
Advanced crypto implementation	28/05/2026	05/09/2026	100
Project publication on platforms	06/09/2026	12/09/2026	6

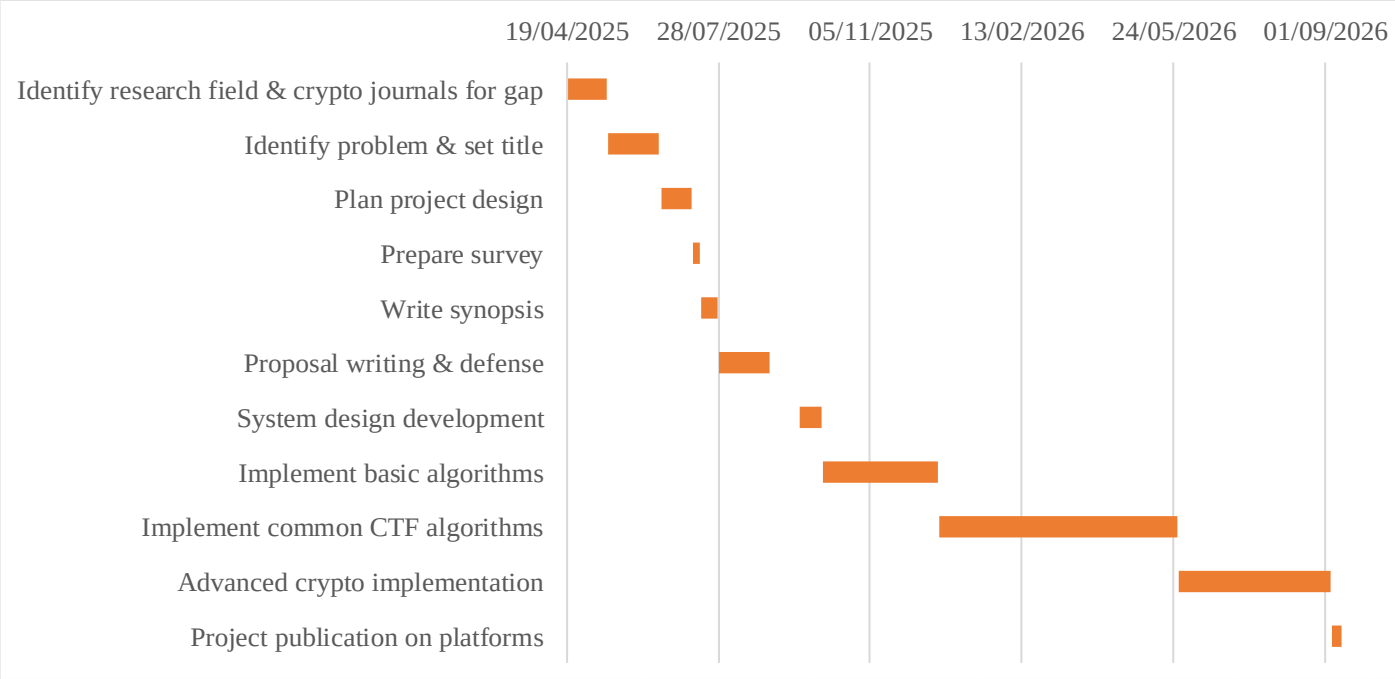

Figure 3: Project activities

6. BUDGET

SOKONALYSIS CIPHER TOOKIT EXPECTED BUDJECT					
Item	**Quantity**	**Days**	**Pages**	**Cost (K)**	**Total (K)**
Laptop (already available)	2			0	0
Internet (research/testing)	1	482		5	2410
Flash Drive (backup)	1			0	0
Printing	1		23	2.5	57.5
Binding	1			15	15
Software Publication (free)	2			0	0
					2482.5

Table 1: Project budget

Year	**Source of Income (K)**		**Total (K)**	**EXPECTED BUDGET (K)**
	Personal	**HELSB**		
2025	741.25	0	741.25	2482.5
2026	741.25	1000	1741.25	
			2482.5	

Table 2: Project financial feasibility

7. Appendices

7.1 Appendix A: Survey Questionnaire

Microsoft Form Response

14. What difficulties do you face when solving crypto-related CTF challenges?

Figure 4: Difficulties faced during Crypto CTF challenges

15. Would you like to use a tool that can help you solve Cryptography CTF challenges in one place?

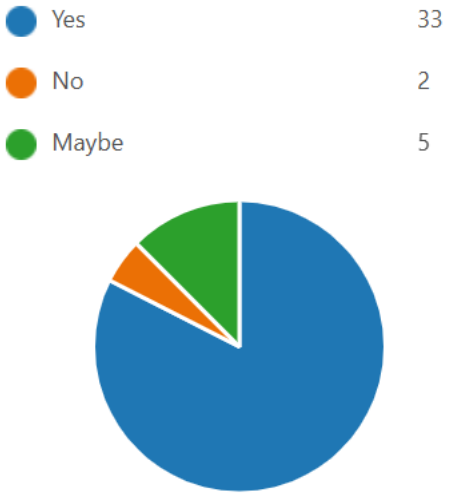

Figure 5: Need for a Crypto CTF focused tool